

$$J_N = \frac{I_N}{A}$$

$$J_P = \frac{I_P}{A}$$

$$I_N = J_N A$$

$$I_P = J_P \times A$$

electron current

hole current.

MODULE III

TRANSISTORS

(transfer of resistance)

Bipolar Junction transistors

transistors

Junction transistor es: (BJT)

Field effect transistors (FET)

Junction field effect (JFET)

Metal oxide semiconductor field effect T (MOSFET)

→ BJT (Bipolar junction transistor)

* It is called Bipolar because it has electron current and hole current (I_n, I_p)

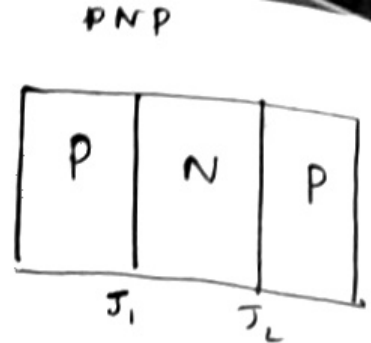
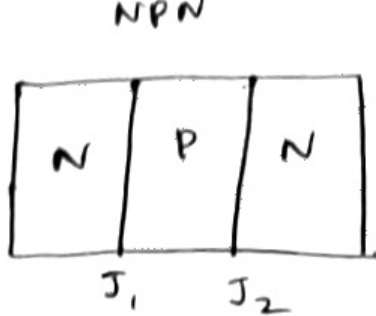


* here there are two junctions



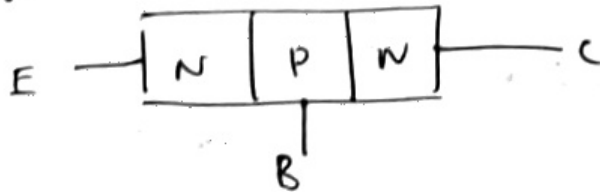
* BJT → PNP
→ NPN

two types of BJT transistors PNP, NPN

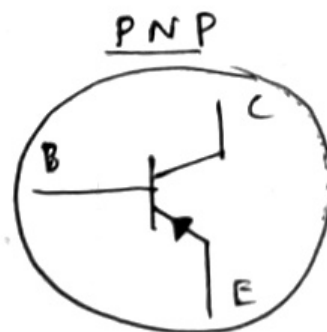
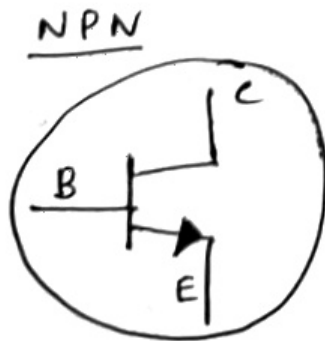


* transistor it has 3 terminals they are emitter, base, collector

ex:



* symbol



current flow
+ve $\rightarrow$ -ve

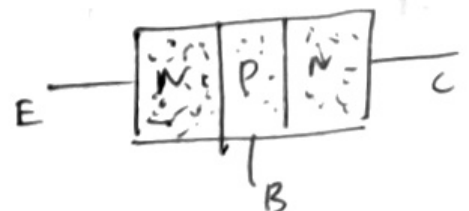
$\rightarrow$ Doping and area profile of a transistor

* Emitter:- doping

heavily doped

* Moderately doped collector

* lightly doped Base



Area

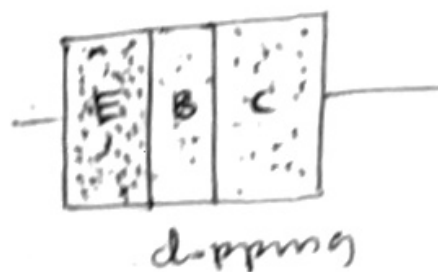
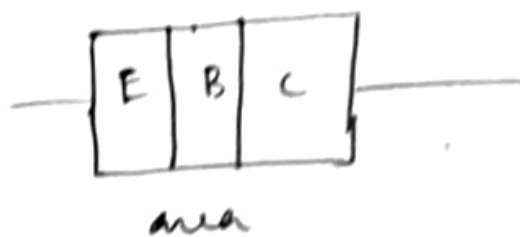
* emitter has medium area

* collector have max area such that it collects the charge carriers

* base minimum area

- emitter is heavily doped because emitter is emitting charge carriers
- Base through which charge carriers are passing, it does not collect or emit e^- or h^+ , less area.
- collector it will collect the emitted charge carriers and it has max area for collecting

Parameter	Emitter	Collector	Base
Doping	heavily doped	moderately doped	lightly doped
Area	moderate area	maximum area	minimum area



Regions of operations

a transistor normally works in three regions

(i) cut off region

(ii) saturation region

(iii) Linear (follow ohm's law)

also called ohmic region or resistive region or triode region.

NB

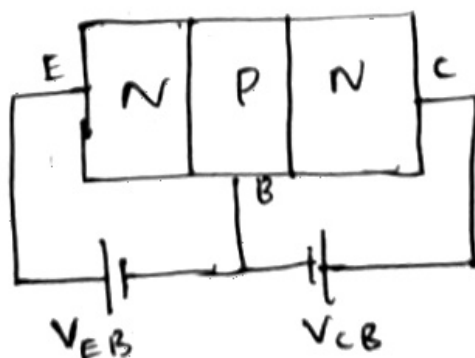
Regions of operation

region	Emitter Base	collector Base
cut off	Reverse bias (RB)	RB
saturation	Forward bias (FB)	FB
Linear / active Reverse region	FB	RB
inverse active	RB	FB

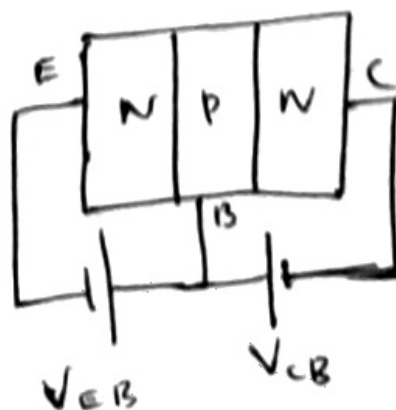
* transistor can work as a switch
when it is in cut off region i.e. both
emitter base and collector base in reverse
bias

→ NPN

* NPN transistor in cut off region.



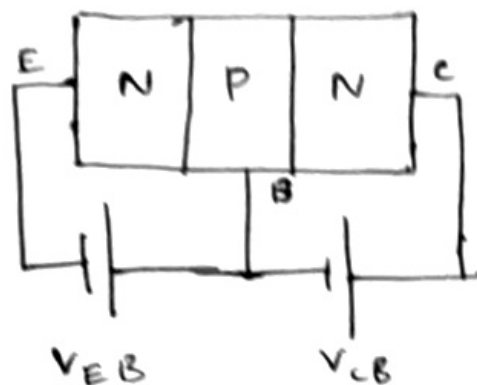
* NPN transistor in saturation region



drawn according
to the figure

*

NPN transistor in linear or resistive region



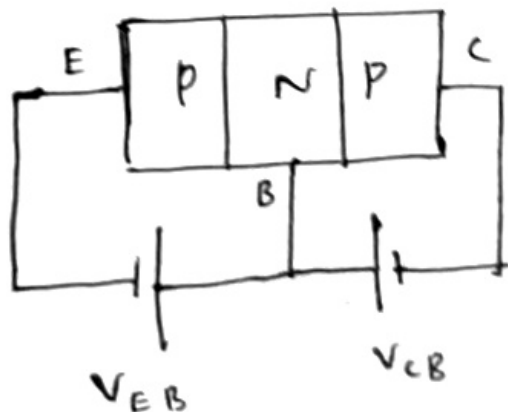
→

PNP

Regions of operation of a PNP transistor

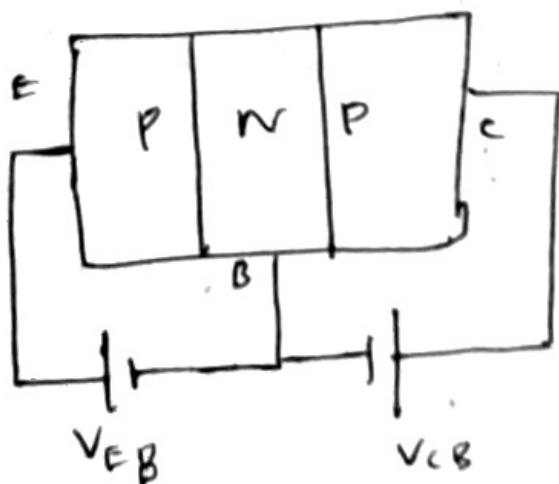
*

PNP in cutoff

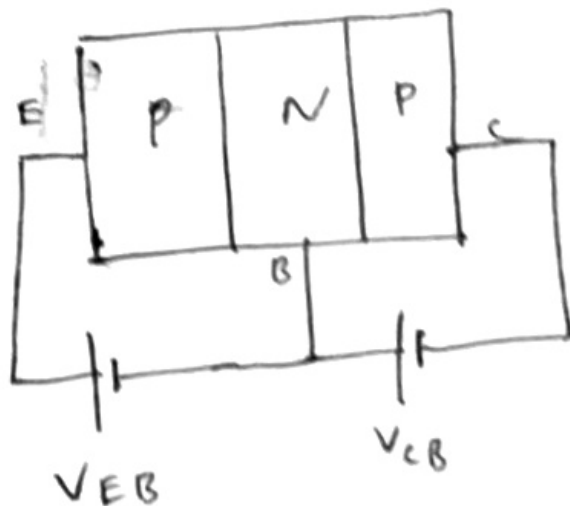


*

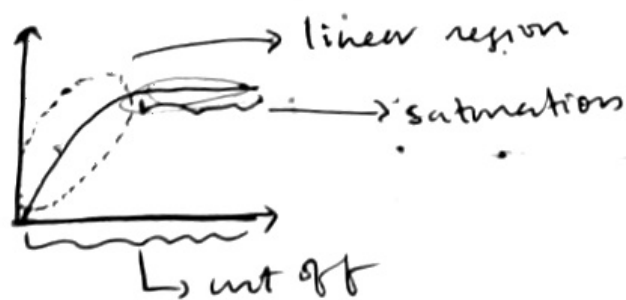
PNP in saturation



* PNP in linear



→ Major application of transistor is in Linear region eg: Amplifier



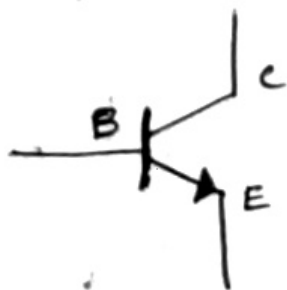
→ Another region

Inverse active region (Reverse of linear region)
 emitter base output in Reverse bias (RB)
 collector base " in Forward bias (FB)

EB → RB
 CB → FB

→ Transistor Configurations

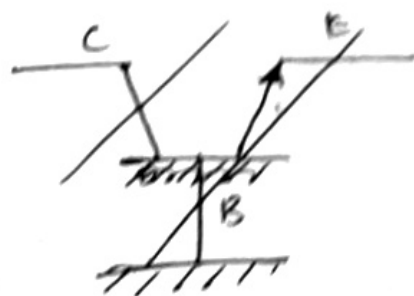
NPN



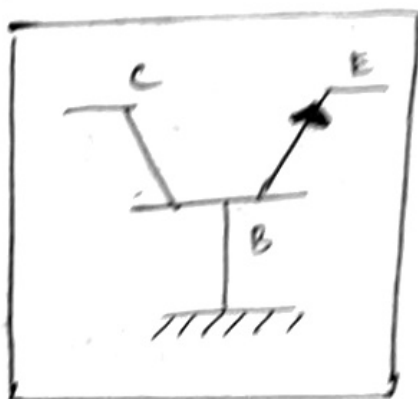
There should be four terminals as two terminal for input and two terminals for output
 ∴ one terminal out of three we are making as common terminal
 this is called transistor configuration
 eg: common base (CB), CE, CC

(NB)
 ommp

Common Base (NPN)



here base is grounded



here
input $\rightarrow$ EB
output $\rightarrow$ CB

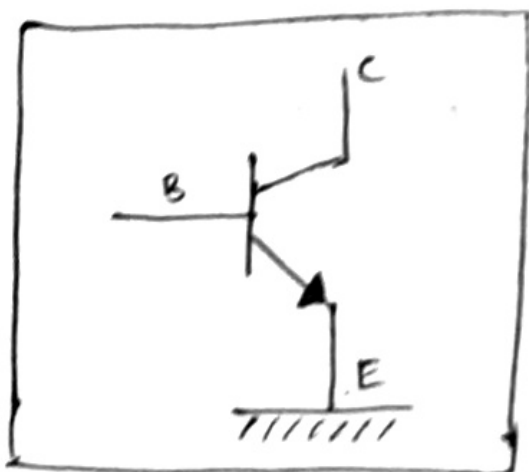
here input is across emitter and base and
output is across collector base and
input current is called I_E and output
current I_C (here transistor works as voltage buffer)

current gain in common Base $= \alpha$

$$\alpha = I_C / I_E$$

$\alpha = 0 - 1$
no amplification

Common Emitter (NPN)



input across EB

output across = CE

input current $= I_B$

output current $= I_C$

current gain in common Emitter

$$\beta = I_C / I_B$$

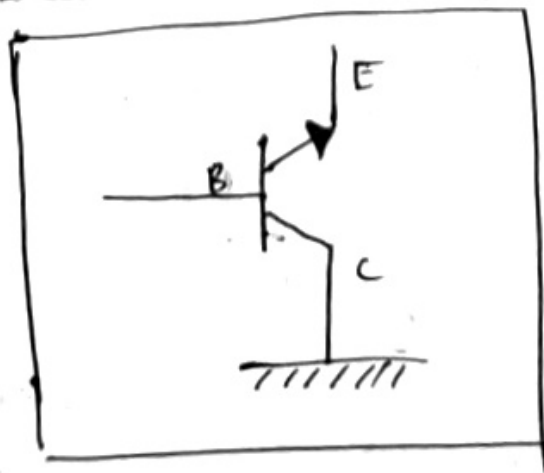
$$\beta = I_C / I_B$$

$$\therefore \boxed{I_C = \beta I_B}$$

$\beta = 20 - 200$
Amplification happens

here works as
 $\therefore$ Transistor as an amplifier
working in common emitter configuration.
because here current amplifies
current obtained is greater than input.

* common collector (NPN)



input across CB
output across CE
input current = I_B
output current = I_E

(2) current gain in common collector

$$\boxed{\gamma = \frac{I_E}{I_B}}$$

in this region transistor works as a
constant current source.

→ Relationship between α , β , β

$$I_E = I_C + I_B$$

$$\alpha_{dc} = \frac{I_C}{I_E}$$

$$\alpha_{ac} = \frac{I_C}{I_E}$$

$$\beta_{dc} = \frac{I_C}{I_B}$$

$$\beta_{ac} = \frac{I_E}{I_B}$$

$\alpha_{dc} = \frac{\beta}{\beta + 1}$	$\beta_{dc} = \frac{\alpha}{1 - \alpha}$	$\beta_{ac} = \beta + 1$
---	--	--------------------------

NB (Q).

For an NPN transistor $\alpha_{dc} = 0.98$
Base current (I_B) = 100 μ A find β , I_C , I_E ?

$$\alpha = \frac{I_C}{I_E} \quad \alpha = \frac{\beta}{\beta + 1} \quad \beta = \frac{\alpha}{1 - \alpha}$$

$$\beta = \frac{0.98}{1 - 0.98} = \frac{0.98}{0.02} = 49$$

$$\beta = \frac{I_C}{I_B} \Rightarrow \frac{49}{100}$$

$$49 = \frac{I_C}{100}, I_C = 4900 \mu A$$

$$\alpha = \frac{I_C}{I_E} \Rightarrow \frac{4900}{I_E} = \frac{100}{4900} \Rightarrow I_E = \frac{100}{0.02} = 5000 \mu A$$

NB

comparison of transistor configuration (based on area of domain)

parameter	CB	CE	CC
input impedance (Z)	very low	low	high
output Z	very high	medium	low
$\frac{I_o}{I_i}$ current gain (A _I)	α less than 1 $I > \alpha C$	high (β)	very high ($\beta + 1$)
$\frac{V_o}{V_i}$ voltage gain (A _V)	high	high	low
use	current source	Amplifier	Voltage buffer

Transistor characteristics

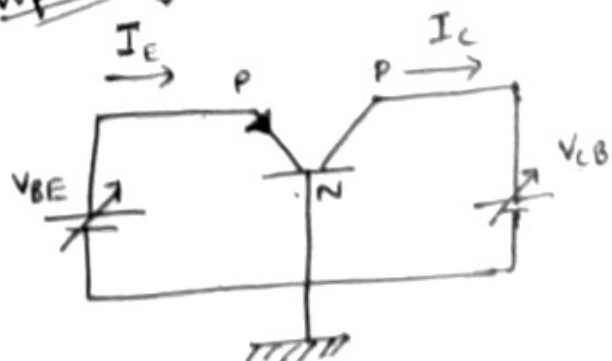
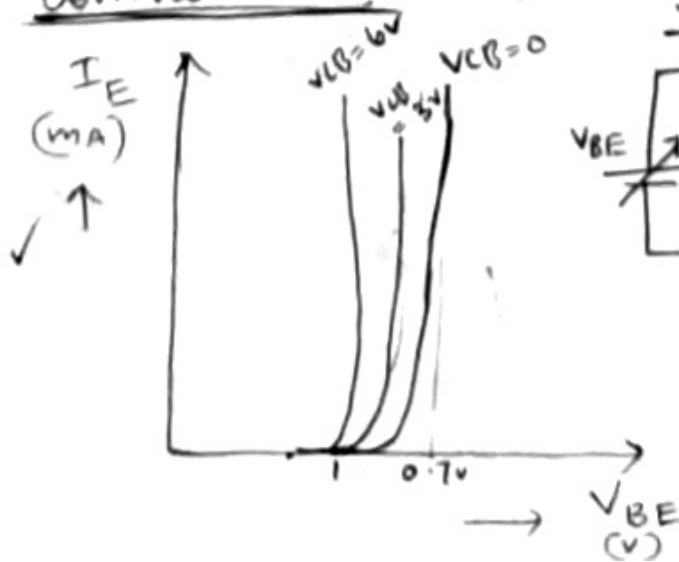
input characteristics

plot b/w input voltage and input current for different values of output voltage

output characteristics

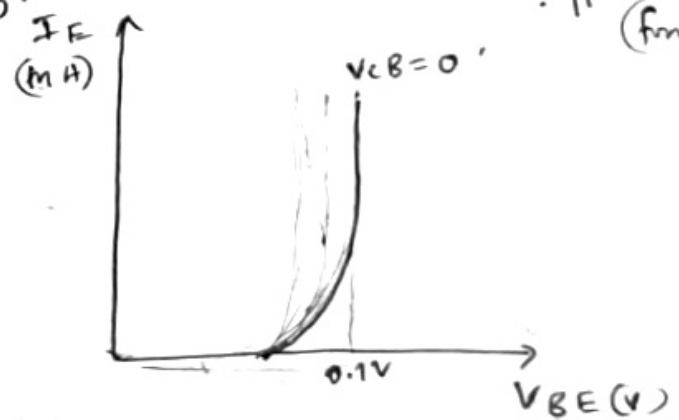
plot b/w output voltage & output current for different values of input current

common base (CB) (input characteristics)

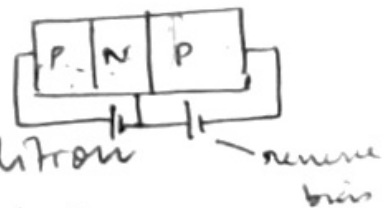


$$I_E = I_C + I_B$$

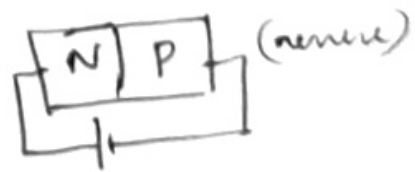
when $V_{CB} = 0$ the emitter and base portion of the transistor behaves as a forward biased PN junction. then we obtain:



if we increase reverse bias the size of depletion region will also increase.

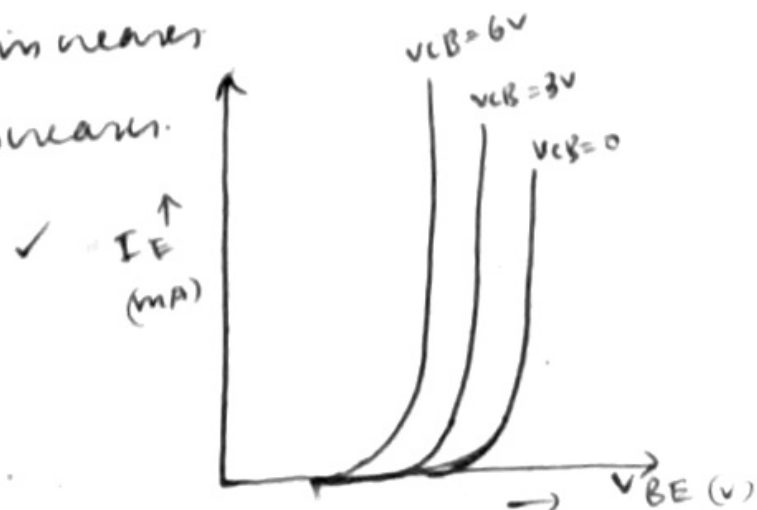


Early effect:- in reverse biased condition the base current is reducing while other current is increasing as the depletion base is lightly doped thus depletion region increases, towards base.

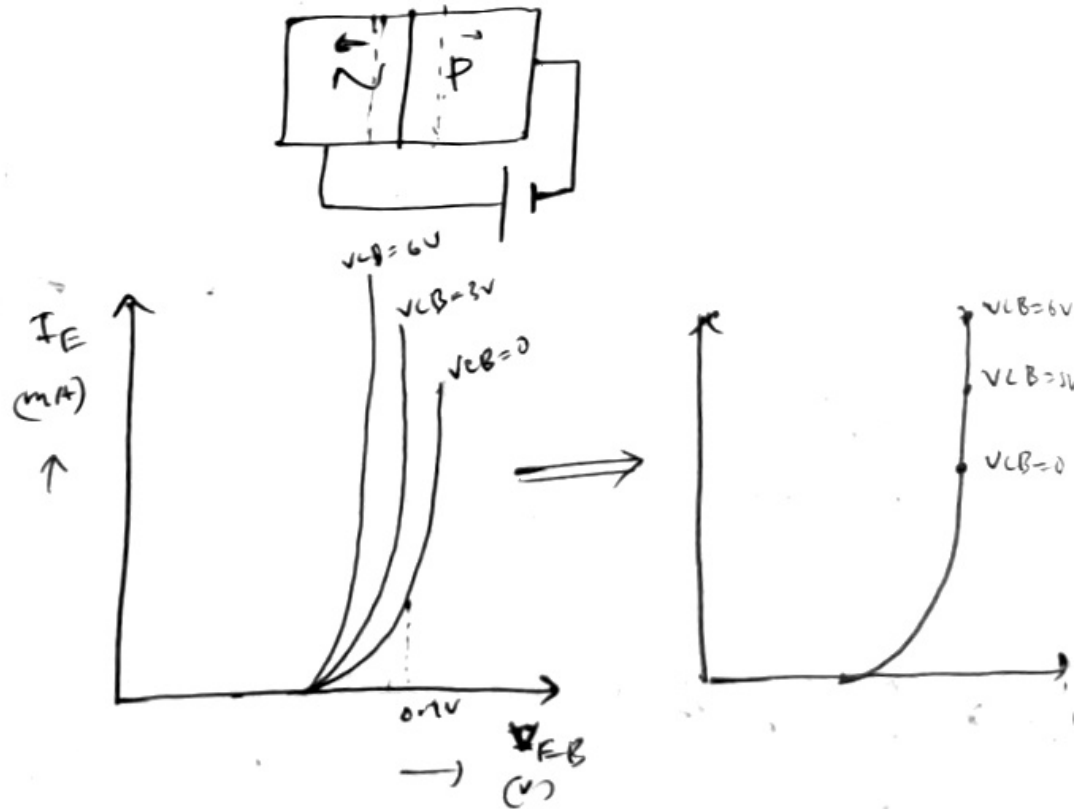


effects

- * Base current ~~increases~~ decreases
- * emitter current increases
- * collector current increases.

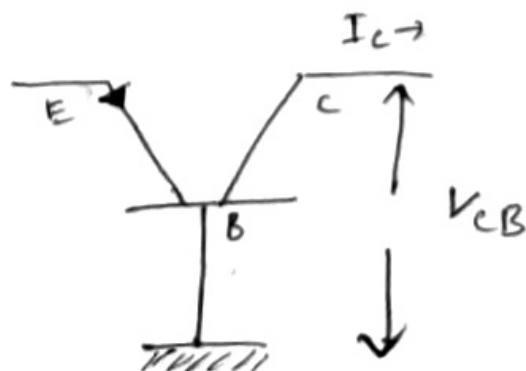


→ Early effect
 since the base is lightly doped
 the penetration of depletion region
 towards base is more than collector.



→ Output characteristics
common base (CB)

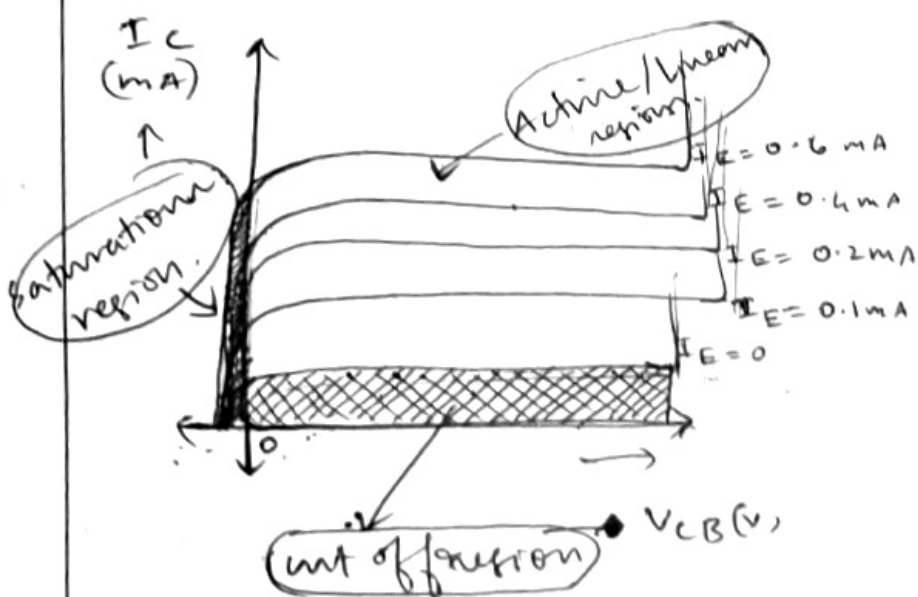
plot b/w output v and op output I
 for different values of input current



* here output voltage is taken in reverse bias PN Junction.



* I_E is input current, when I_E is input region is reverse biased, $I_E = 0$ at that condition both input and output region is in reverse bias. Then this region is called int off region.



CB		Region
EB	CB	int off
R	R	
EB	CB	Active
F	R	region
EB	CB	Saturation
F	F	region

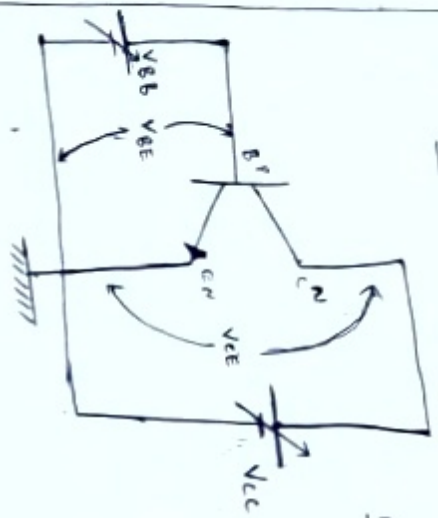
* after int off region an active region/resistive/linear/forward region is formed while EB Junction is forward biased and CB Junction is reverse biased.

* after when both Junction are in forward bias a saturation region is formed after 0 in the V_{CB} .

The shape of the curve

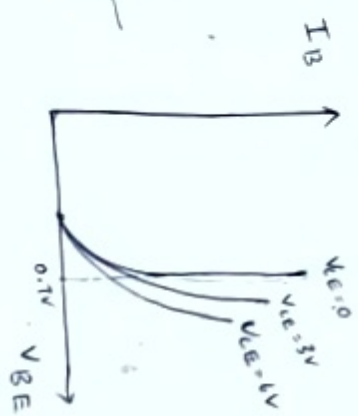
common

NPN transistor



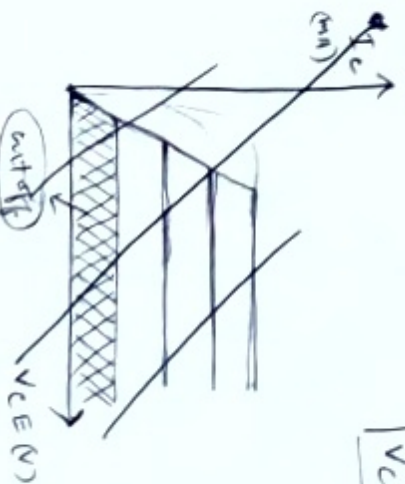
input forward bias
output reverse bias

input characteristics (CE)



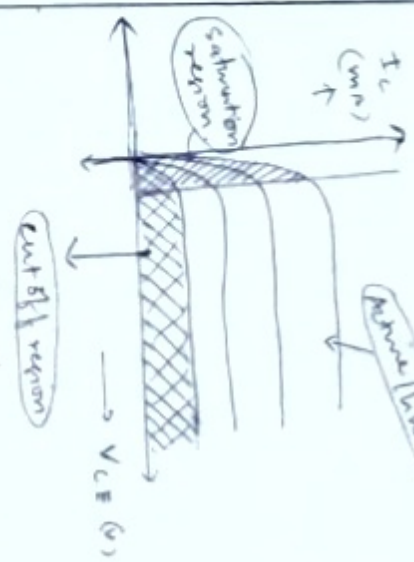
due to Early effect
base current reduces
emitter current and
collector current
increases
 $I_B \uparrow \Rightarrow I_E \uparrow \Rightarrow I_C \uparrow$

output characteristics (CE)



$$V_{CE} = V_{BE} + V_{CB}$$

reverse bias



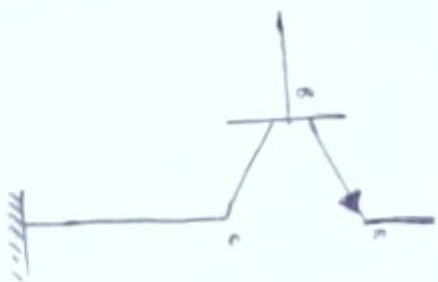
$$V_{CE} = V_{BE} + V_{CB}$$

of the value V_{CE} the 0 beyond origin we
determine saturation region

let here when V_{CE} becomes -ve then
 V_{CB} also becomes -ve by the eqn $V_{CE} = V_{BE} + V_{CB}$

$$-V_{CE} = -V_{BE} - V_{CB}$$

common collector characteristics (CC)



Source regulation

- * AC supply in the transformer (input AC)
- * AC input $\pm 10\%$
- * $\text{Pfl} \Rightarrow \text{Pfl}$
- * to eliminate the variation in the input AC

Load regulation

- * output current (I_L) or load current
- * eliminating ΔE
- * only E eliminating the additional voltage due to fluctuation in source

Source effect defines or source regulation.

how the output voltage (E) change by a factor (ΔE)

due to change in input voltage of the output voltage changes 100 mV then the input AC voltage changes by 10% then source effect is 100 mV

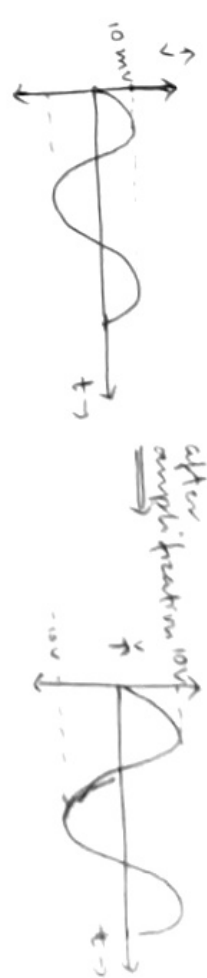
Source effect = ΔE for 10% change in input voltage.

Load effect defines or load regulation

how the output voltage changes when the load current is increased from 0 $\rightarrow$ maximum value.

load effect = ΔE for load current (I_L)

Transistor as Amplifier

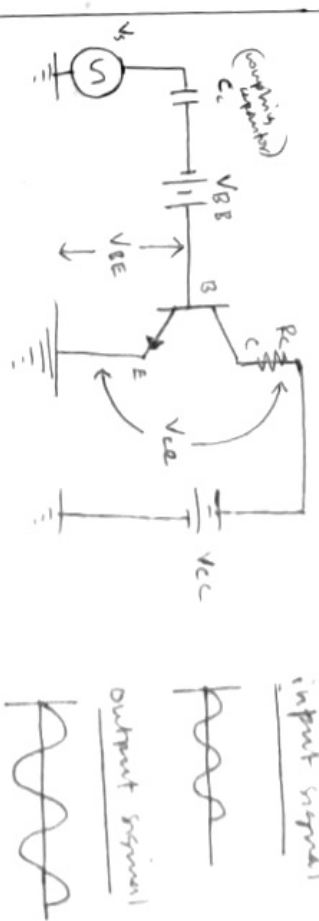


amplification factor (A_v) = $\frac{V_o}{V_i} = \frac{10V}{10mV} = 1000$

practical unit of amplification is dB

NB

How transistor work as amplifier configuration. (fig telling the working)



$$\beta = \frac{I_c}{I_b} = \frac{I_c}{I_b}$$

ie output current $I_c = \beta$ times I_b

$$I_c = \beta I_b$$

by connecting a resistor after collector ie R_c & output is $V_o = V_c = I_c \times R_c$

($\beta = 20 \rightarrow 200$)

by approximating base current = input current

$$V_i = V_{be} = I_e \times r_e$$

internal resistance of emitter (r_e)

neglecting base current

$$\frac{V_o}{V_i} = \frac{I_c \times R_c}{I_e \times r_e} = \frac{R_c}{r_e}$$

$$\boxed{\frac{V_o}{V_i} = \frac{R_c}{r_e}}$$

$$\boxed{I_c = I_b + I_e}$$

$$I_b \approx 0$$

$$\therefore I_c = I_e$$

NB
→

Transistor as Switch

blw cut off region and saturation region
transistor work as switch.

when transistor is on - saturation region.

emitter base - Forward

collector base - F B 

when transistor is off - cut off region

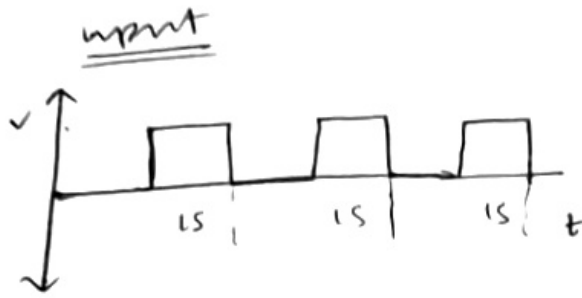
emitter base - reverse

collector base - reverse



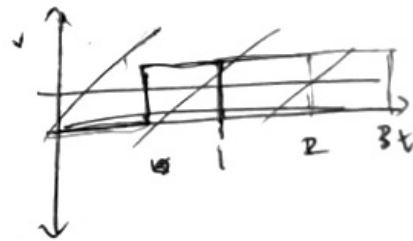
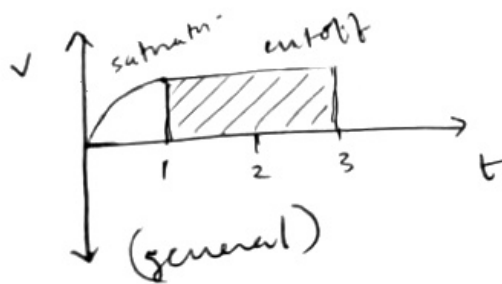
Q) A square wave is given input to a transistor and it is off for 2s and on for 1s draw output

off - 2s
on - 1s



off - cutoff 2s
on - saturation - 1s

output



output graph



saturation -
cutoff region.

so here transistor

